

Ivan Doubovik

Laboratoire Paul Painlevé

Université de Lille

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French and United States citizen

Education

PhD in Mathematics 2023–Present

Université de Lille

Supervisors: Nicole Raulf and Didier Lesesvre

Dissertation: Quantum Unique Ergodicity for Hilbert Modular Forms

Master of Mathematics 2021–2023

Université de Lille

Bachelor of Mathematics 2019–2021

Université de Lille

Bachelor of Physics 2015–2018

Université de Lille

Teaching Experience

Université de Lille

- 2025–2026: Advanced Linear Algebra (second-year mathematics students), problem sessions (36 h).
- 2024–2025: Mathematics (first-year undergraduate students: mathematics and computer science), problem sessions (15.5 h).
- 2024–2025: Mathematics for Biology (first-year biology students), problem sessions (16.5 h).
- 2023–2024: Mathematics (first-year undergraduate students: mathematics and computer science), problem sessions (32 h).

Academic Activities

Number Theory

- Organized a reading group (2025) on automata-theoretic approaches in number theory based on the paper *Automata in Number Theory* by Boris Adamczewski.
- Presented results from my PhD research on quantum unique ergodicity at the Aarhus Conference on Automorphic Forms (2025).
- Summer School: Arithmetic Geometry and Analytic Number Theory, CIRM Marseille (2024).
- Conference: Automorphic Forms and L -functions, Queen Mary University of London (2023).
- ArStAFANT Workshop, EPFL (2023).

Logic and Foundations

- Workshop *Types and Topology*, University of Birmingham (2025).
- GT Scalp Workshop, Créteil–Vitry (2025).
- MALINCA Kickoff Meeting, IHP Paris (2025).

Research Experience

Second year master's Research Project (2023)

QUE for shrinking sets under Nicole Raulf and Didier Lesesvre.

Master's Research Project (2022)

Tate's Thesis under Didier Lesesvre.

Research Internship (2018)

Quantum optomechanics, Université de Lille.

Independent Study in Logic and Type Theory

- Studied *Introduction to Homotopy Type Theory* by Egbert Rijke.
- Studied *Basic Proof Theory* by Troelstra and Schwichtenberg.
- Reading introductory material on normalization and canonicity in dependent type theory.
- Learning the Lean 4 proof assistant.
- Regular participant in the Formath seminar (IRIF, Paris), focused on formalizing mathematics.

Technical Skills

- Lean 4 (basic, currently learning)
- $\text{\LaTeX}$
- C++
- OpenGL
- Languages: English (fluent), French (fluent)

References

Nicole Raulf

Université de Lille

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Didier Lesesvre

Université de Lille

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